Systematic Review and Meta-Analysis

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Comparative efficacy of ibuprofen and low-level laser therapy on pain intensity after elastomeric separator placement

A systematic review and network meta-analysis

Shasha Gao, BD[a], Peipei Wu, BD[b], Yang Cheng, BD[b,*]

Abstract

Background: This research aimed to assess the efficacy of ibuprofen and low-level laser therapy (LLLT) in pain reduction following the placement of elastomeric separators, utilizing a network meta-analysis (NMA) to synthesize data from relevant randomized controlled trials (RCTs). Our objective was to determine the most effective method for pain management in orthodontic patients.

Methods: This study was registered with International Prospective Register of Systematic Reviews (PROSPERO) (registration number: CRD42025640519). A comprehensive literature search was conducted across various databases up to January 19, 2025, to identify RCTs evaluating the impact of ibuprofen and LLLT on pain intensity following orthodontic separator placement. The quality of the included studies was assessed using the Cochrane risk-of-bias tool. A NMA was performed to combine direct and indirect comparisons, ranking the interventions based on their effectiveness in alleviating pain at 6 hours and 24 hours after orthodontic separator placement.

Results: Twenty-three studies with 1007 participants were included in the NMA. Both ibuprofen and LLLT demonstrated significantly greater reductions in pain intensity compared to control or placebo groups at all assessed time points. For ibuprofen, pain reduction was observed at 6-hour with a mean difference (MD) of –1.93 and a 95% confidence interval (CI) of –2.93 to –0.93, and at 24-hour with an MD of –1.63 (95% CI: –2.58 to –0.68). For LLLT, the MD at 6-hour was –1.46 (95% CI: –2.28 to –0.64), and at 24-hour, the MD was –1.84 (95% CI: –2.52 to –1.16). Ibuprofen showed greater effectiveness in reducing pain intensity at 6-hour, with a surface under the cumulative ranking curve (SUCRA) of 88.4, whereas LLLT demonstrated more sustained reduction at 24-hour, with a SUCRA of 81.8.

Conclusion: This NMA provides evidence that both ibuprofen and LLLT are effective in reducing pain following elastomeric separator placement. Ibuprofen offers more immediate pain reduction, whereas LLLT offers more sustained effects over 24-hour. These findings suggest that a time-dependent, multimodal approach may be beneficial for optimizing orthodontic pain management. To further validate these findings and establish standardized clinical protocols, more high-quality randomized clinical trials are needed, particularly those involving direct, head-to-head comparisons between ibuprofen and LLLT.

Abbreviations: CI = confidence interval, GRADE = grades of recommendations, assessment, development, and evaluation, IQR = interquartile ranges, LLLT = low-level laser therapy, MD = mean difference, NMA = network meta-analysis, NSAIDs = nonsteroidal anti-inflammatory drugs, OHIP-14 = oral health impact profile-14, OHRQoL = oral health-related quality of life, PrBest = probability of being the best treatment, PRISMA = preferred reporting items for systematic reviews and meta-analyses, PROSPERO = international prospective register of systematic reviews, RCTs = randomized controlled trials, SD = standard deviations, SUCRA = surface under the cumulative ranking curve, VAS = visual analog scale.

Keywords: elastomeric separator, ibuprofen, low-level laser therapy, network meta-analysis, pain intensity

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The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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1. Introduction

Orthodontic pain, resulting from diverse treatment procedures including separator placement, initial wire engagement, banding, the use of elastics, rapid maxillary expansion, and debonding, can substantially affect quality of life by impairing chewing, speech, and daily activities, potentially reducing treatment compliance or even leading to early discontinuation.[1–5] Numerous studies have shown that orthodontic pain significantly impairs oral health-related quality of life (OHRQoL), particularly in domains such as physical pain and psychological discomfort.[6] Clinical and cross-sectional investigations using validated tools such as the oral health impact profile-14 (OHIP-14) consistently demonstrate that patients with higher pain scores tend to report significantly poorer OHRQoL outcomes.[7,8] However, the prevalence and severity of orthodontic pain are influenced by a variety of factors, including age, sex, and treatment modality. For instance, in a cohort of 160 patients, 58.3% of adults reported severe pain with visual analog scale (VAS) scores between 8 and 10, whereas children aged 6 to 12 years mostly experienced mild or no pain. Female patients have also been reported to show increased pain sensitivity and greater use of analgesics, possibly due to hormonal and psychosocial factors.[9] In a 6-month prospective cohort study, Chan et al observed that 61% to 83% of patients undergoing treatment with fixed appliances reported pain during multiple adjustment appointments, in contrast to 18% to 27% of those using clear aligners. Among all orthodontic procedures,[10] the application of separators is acknowledged as one of the most insufferable procedures.[11,12] These separators are utilized to generate interproximal space between molars, facilitating the placement of orthodontic bands.[13] Nevertheless, their application results in the stretching or compression of the periodontal ligament, subsequently provoking an inflammatory response.[14,15] Research has shown that the pain associated with elastomeric separators generally begins approximately 6-hour after placement, peaks in intensity at 24-hour, and gradually subsides over a span of 5 days.[16] Consequently, the development of effective pain alleviation strategies is crucial for enhancing patient compliance and ensuring the successful completion of orthodontic treatment.

To mitigate orthodontic pain, both pharmacological and non-pharmacological strategies have been investigated, with nonsteroidal anti-inflammatory drugs (NSAIDs) and low-level laser therapy (LLLT) being among the most extensively studied methods.[17] Ibuprofen, a widely used NSAID, exerts its analgesic effects primarily through the inhibition of cyclooxygenase enzymes, leading to a reduction in prostaglandin synthesis and subsequent attenuation of inflammation-induced pain.[18] Despite its broad availability and well-documented efficacy, ibuprofen is associated with certain limitations, such as potential gastrointestinal side effects and the possible deceleration of orthodontic tooth movement and periodontal tissue remodeling.[19–22] On the other hand, LLLT represents a noninvasive therapeutic modality that can enhance local blood circulation, stimulate ATP synthesis, diminish the release of inflammatory mediators, augment osteoclast and osteoblast activity, and provide analgesic benefits.[23–25] Although LLLT is generally well tolerated and has demonstrated some effectiveness in pain management, the variability in clinical outcomes across studies presents significant challenges for its widespread implementation.[26]

Although both ibuprofen and LLLT have demonstrated effectiveness in managing orthodontic pain, a definitive consensus regarding their relative efficacy remains elusive. Prior systematic reviews have predominantly concentrated on either pharmacological or non-pharmacological interventions independently, with a limited number of studies conducting direct comparisons between these 2 approaches.[27] Additionally, variations in study designs, patient populations, and pain assessment methodologies complicate direct comparisons between studies. In light of these limitations, a network meta-analysis (NMA) is justified. Unlike traditional pairwise meta-analyses, NMA facilitates the integration of both direct and indirect evidence,[28–30] allowing for a comparative ranking of ibuprofen and LLLT in terms of their efficacy on pain intensity following orthodontic separator placement. Therefore, this study aimed to determine, in patients undergoing elastomeric separator placement, whether ibuprofen or LLLT is more effective in reducing pain intensity compared with each other or with control interventions, at 6 and 24 hours post-placement, based on evidence derived from randomized controlled trials (RCTs).

2. Methods

2.1. Design and registration

Initially, a scoping search was conducted in PubMed to assess the availability of existing systematic reviews and identify related studies potentially eligible for inclusion. A significant number of studies directly relevant to the topic of this review were found in the preliminary search. Consequently, we proceeded to develop a detailed study protocol, which was prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO) under registration number CRD42025640519. This systematic review adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension guidelines for reporting systematic reviews that incorporate network meta-analyses of healthcare interventions.[31,32] The meta-analysis utilizes previously published data, so ethical approval was not required.

2.2. Eligibility criteria

2.2.1. Inclusion criteria. Studies were included if they enrolled patients undergoing orthodontic treatment with elastomeric separators, regardless of age, sex, ethnicity, or socioeconomic background. Eligible studies had at least 1 intervention group receiving ibuprofen or LLLT for pain management, without restrictions on laser parameters such as wavelength, energy density, power output, exposure duration, treatment intervals, or follow-up period. The comparison group could receive a placebo, no treatment, or any alternative intervention distinct from the experimental group. To ensure consistency in outcome measurement, only studies that assessed pain using the VAS as a primary measure were included. The review was limited to RCTs published in English.

2.2.2. Exclusion criteria. Studies were excluded if they involved participants with chronic pain conditions, such as temporomandibular joint disorder, fibromyalgia, chronic migraine, or other persistent pain syndromes. Studies were also excluded if participants had a history of neurological, psychiatric, or cognitive conditions. In addition, we excluded studies not involving orthodontic separators, those in which the intervention did not include ibuprofen or LLLT, or those that did not evaluate pain using VAS. Nonrandomized studies, reviews, meta-analyses, case reports, case series, in vitro studies, conference abstracts, publications not in English, and duplicate reports were also excluded.

2.3. Search strategy and study selection

We carried out a comprehensive electronic search for eligible trials in PubMed, Web of Science, Cochrane, EMBASE and the English Clinical Trial Registry. The search covered studies published from January 2000 to January 19, 2025, and was limited to English-language publications to ensure clinical relevance, methodological consistency, and technological comparability. Search strategies were constructed based on the PICOS (population, intervention, comparator, outcome, study design)

framework and tailored to the syntax requirements of each database. A combination of Medical Subject Headings (MeSH)/Emtree terms and free-text keywords was used. Boolean operators were applied to enhance sensitivity ("OR") and precision ("AND"). The search terms included combinations of concepts related to orthodontic separation procedures (e.g., "separator placement," "orthodontic forces"), pain measurement ("visual analog scale," "VAS"), interventions (e.g., "Ibuprofen," "Low-Level Laser Therapy"), and study design ("randomized controlled trial," "placebo"). Full search strategies for all databases, including keyword combinations and number of records retrieved per step, are provided in Table S1, Supplemental Digital Content, https://links.lww.com/MD/P535.

Studies were included based on predefined eligibility criteria, independently assessed by 2 reviewers (G.S and W.P). Titles and abstracts were initially screened, and then full texts were retrieved for further assessment of studies that might be eligible. Conflicts were resolved through group discussion. For study where the complete text was unavailable or the data were incomplete – such as those presenting results only in graphical form without numerical values – the authors were contacted through email. If they were unreachable after 3 attempts or if the studies were confirmed to be inaccessible, the publications were excluded. The full texts of all potentially relevant studies were subsequently rescreened in duplicate.

2.4. Data collection and quality assessment

The data to be extracted from each trial included author information, publication date, study design, sample size, mean age, interventions, timing of pain assessments, VAS scores, and other relevant data.

The Cochran risk-of-bias tool was employed by 2 reviewers to evaluate the risk-of-bias in each study. This assessment includes 7 domains: random sequence generation (selection bias), allocation concealment (selection bias), blinding of participants and personnel (performance bias), blinding of outcome assessment (detection bias), incomplete outcome data (attrition bias), selective reporting (reporting bias) and other bias. Bias levels was categorized as "Low risk" or "Unclear risk" or "High risk." A domain was rated as "Low risk" if sufficient methodological details were reported and the implementation was judged to be adequate to minimize bias. It was judged as "High risk" if there were clear methodological flaws or if the study lacked critical safeguards, such as absence of blinding in subjective outcomes or high dropout rates without proper handling. "Unclear risk" was assigned when the study did not provide enough information to allow for a definitive judgment regarding that domain.

In addition, the overall quality of evidence for each outcome was assessed using the GRADE (grades of recommendations, assessment, development, and evaluation) system.[33] This approach considers 5 domains: risk-of-bias, inconsistency, indirectness, imprecision, and publication bias or other factors. Based on these criteria, the quality of evidence was graded as high, moderate, low, or very low, and a summary of findings table was provided. Any discrepancies were resolved through consultation with a third assessor.

2.5. Data synthesis and analysis

In this NMA, all statistical analyses were conducted using STATA v14.0 and RevMan v5.4, with the VAS scores serving as the continuous outcome measure. When a study reported pain intensity using a 100-mm VAS, the scores were converted to a 10-point scale by dividing the values by 10, to ensure consistency with other studies. Mean difference (MD) was used as the effect size, and standard deviations (SD) were adjusted using a modified formula assuming a correlation coefficient of 0.5. When original data were presented as medians with interquartile ranges (IQR) or ranges, skewness was assessed using an online tool (www.math.hkbu.edu.hk). If no significant skewness was identified, the data were converted into mean ± SD using established methods.

Pairwise meta-analyses were performed using RevMan v5.4. Statistical heterogeneity among studies was assessed using the I^2 statistic, which quantifies the proportion of variability due to between-study heterogeneity rather than chance. An I^2 value >50%, together with a P-value <.1, was considered indicative of substantial heterogeneity. In such cases, a random-effects model was applied to account for between-study variability; otherwise, a fixed-effects model was used when I^2 was below 50%, suggesting low heterogeneity.

NMA was conducted in STATA v14.0 using a frequentist framework based on a multivariate random-effects meta-regression model. To assess inconsistency between direct and indirect comparisons, the node-splitting method was employed. Additional graphical analyses included network plots (to visualize treatment comparisons), contribution plots (to identify the influence of each comparison), and funnel plots (to explore potential publication bias). Treatment effect rankings were established using surface under the cumulative ranking curve (SUCRA) values and the probability of being the best treatment (PrBest), which reflect both the relative effectiveness and likelihood of optimal intervention performance.

3. Results

3.1. Literature search and selection

The flow diagram presented in Figure 1 delineates the process by which relevant studies were identified. Initially, 297 articles were sourced from 5 databases and manual searches. After removing duplicates, 145 articles proceeded to screening. Title and abstract evaluations narrowed them to 41 articles. Further eligibility checks excluded 18 studies, leaving 23 articles that met the criteria for qualitative synthesis.

3.2. Study characteristics

The studies included in this review are presented in Table 1. All 23 studies were published between 2001 and 2024, and conducted across a diverse range of countries, including Turkey, Iran, India, Brazil, the United States, among others. Out of a total of 1007 patients, 217 were assigned to the ibuprofen group. It is noteworthy that several studies on LLLT utilized a split-mouth control design, which resulted in sample sizes of 568 for the LLLT group and 694 for the control group. Among the studies reviewed, 9 examined the effects of ibuprofen, while 16 focused on LLLT. Furthermore, 2 study directly compared the efficacy of ibuprofen and LLLT following the placement of orthodontic elastomeric separators. Sixteen studies reported VAS scores at 6-hour, all studies reported VAS scores at 24-hour. However, during data extraction, we identified 1 study in which the 24-hour pain scores were presented on a 100-mm scale, despite the methods section stating the use of a 0 to 10 scale. Due to this inconsistency and the lack of clarification regarding the measurement format, the 24-hour data from this study were excluded from the final analysis to maintain methodological consistency.[51] The dose of ibuprofen and the parameters of laser irradiation used within the studies are shown in Tables S2 and S3, Supplemental Digital Content, https://links.lww.com/MD/P535.

3.3. Quality assessment

Among the studies included in the analysis, 5 were identified as high risk-of-bias.[34,38,40,44,51] This was attributed to items such as the blinding of personnel and participants (1/23, 4.35%),[34]

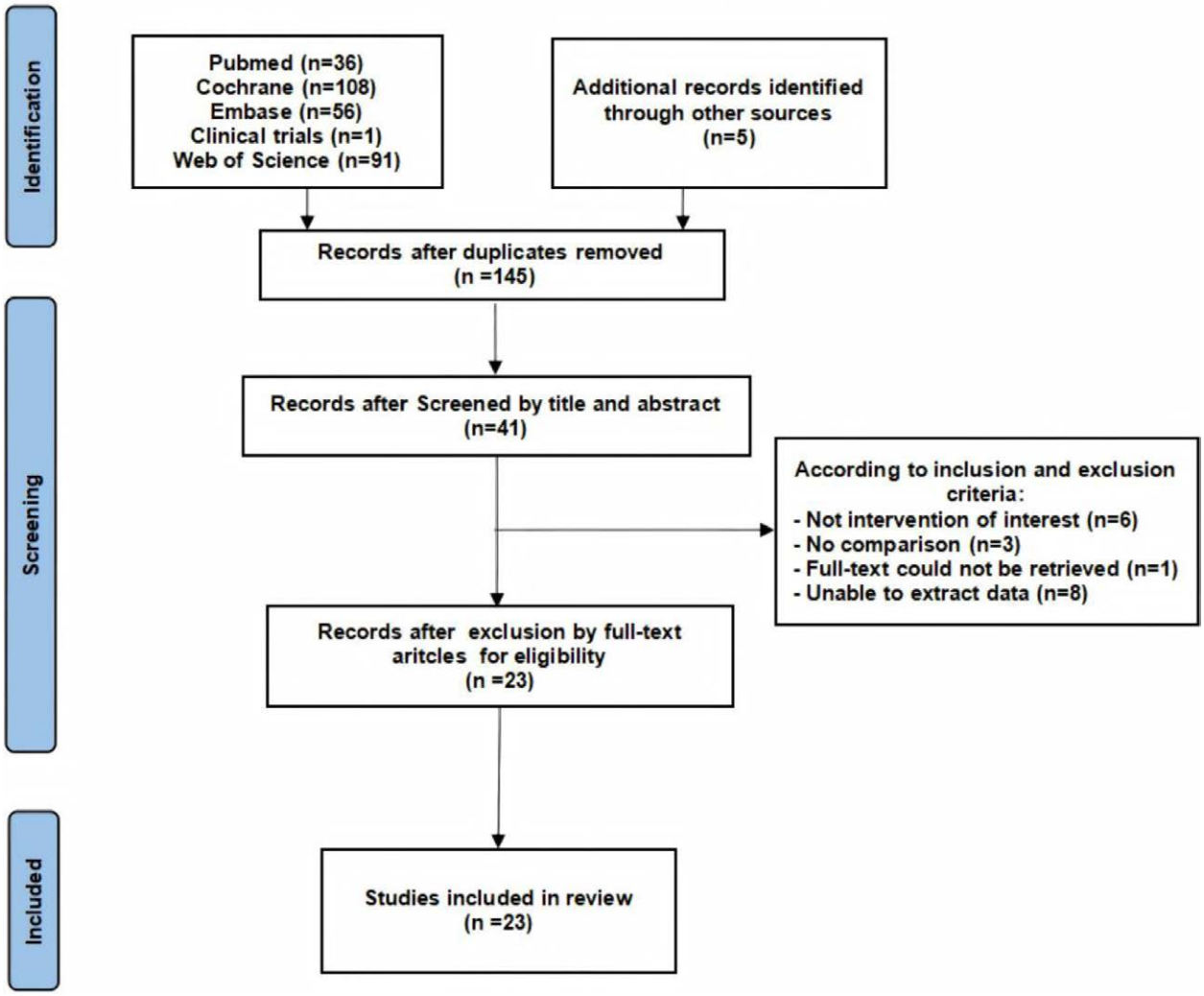


Figure 1. Flow chart of study selection.

incomplete outcome data (3/23, 13.04%),[38,40,44] selective reporting (1/23, 4.35%).[51] Furthermore, 14 studies (14/23, 60.87%) were found to have unclear risk-of-bias,[35–37,41–43,45,47–50,52,55,56] while 4 studies (4/23, 17.39%) were assessed to have a low risk-of-bias (Fig. 2A and B).[39,46,53,54] Detailed judgments and justifications for each risk-of-bias domain are provided in Table S4, Supplemental Digital Content, https://links.lww.com/MD/P535.

According to GRADE, the quality of evidences were very low (Fig. S1, Supplemental Digital Content, https://links.lww.com/MD/P534). These low levels of evidence can be attributed to inconsistencies.

3.4. Traditional meta-analysis results of different interventions

3.4.1. Ibuprofen. The efficacy of ibuprofen in alleviating pain was assessed at 6, 24 hours post-orthodontic treatment. At the 6-hour, 7 studies involving 334 patients revealed significant heterogeneity (I^2 = 90%, P < .00001).[40,46,47,49,51,54,56] A random-effects model demonstrated a significant reduction in pain (MD = −1.88, 95% CI [−3.07 to −0.70], P = .002) (Table 2, Fig. S2A, Supplemental Digital Content, https://links.lww.com/MD/P534). At 24-hour, 7 studies with 376 patients exhibited moderate heterogeneity (I^2 = 72%, P = .002),[40,41,46,47,49,54,56] and the random-effects model indicated a significant reduction in pain levels (MD = −1.43, 95% CI [−2.17 to −0.69], P = .0001) (Table 2, Fig. S3A, Supplemental Digital Content, https://links.lww.com/MD/P534).

3.4.2. LLLT. The effects of LLLT were evaluated at the same time intervals. At 6-hour, 10 studies involving 334 patients revealed substantial heterogeneity (I^2=97%, P < .00001).[34–37,42,44,47,48,52,53] A random-effects model demonstrated a significant reduction in pain (MD = −1.44, 95% CI [−2.42 to −0.46], P = .004) (Table 2, Fig. S2B, Supplemental Digital Content, https://links.lww.com/MD/P534). At 24-hour, 15 studies with 371 patients showed high heterogeneity (I^2=93%, P < .00001),[34–38,42–45,47,48,50,52,53,55] and the random-effects model again indicated a significant reduction in pain (MD = −1.89, 95% CI [−2.71 to −0.06], P < .00001) (Table 2, Fig. S3B, Supplemental Digital Content, https://links.lww.com/MD/P534).

Both ibuprofen and LLLT consistently demonstrated significant pain alleviation at all time points, supporting their clinical application in the management of pain following orthodontic procedures.

3.5. Network meta-analysis

Based on the studies included in this analysis, the article examined outcomes at 6- and 24-hour following the placement of elastomeric separators. The study utilized 2 network graphs, where each node, depicted in green, represents a distinct intervention.

Table 1

Characteristics of included studies.

No.	Study	Country	Design	Sample size	Age (yr)	Intervention	Outcome measures (VAS)
1	Al-Hanbali et al. (2024)[34]	Syria	RCT	E: 48	18–25	E: LLLT	6 h: E: 0.47 ± 0.5/C: 3.88 ± 2.83
				C: 49		C: Control	24 h: E: 1.27 ± 0.95/C: 5.79 ± 2.72
2	Almallah (2016)[35]	Syria	RCT	E: 36	12–26	E: LLLT	6 h: E: 1.84 ± 1.93/C: 3.94 ± 2.8
				C: 36		C: Control	24 h: E: 2.72 ± 2.1/C: 4.18 ± 2.4
3	Almallah (2020)[36]	Syria	RCT	E: 36	15–25	E: LLLT	6 h: E: 1.34 ± 1.94/C: 2.72 ± 2.7
				C: 36		C: Control	24 h: E: 2.26 ± 2.82/C: 3.33 ± 3.17
4	AlSayed (2018)[37]	Syria	RCT	E: 25	16–22	E: LLLT	6 h: E: 0.86 ± 1.27/C: 1.88 ± 2.29
				C: 25		C: Control	24 h: E: 3.51 ± 2.62/C: 4.05 ± 2.3
5	Anicic (2020)[38]	Croatia	RCT	E: 22	12–19	E: LLLT	24 h: E: 3.26 ± 2.25/C: 3.82 ± 2.34
				C: 22		C: Control	
6	Azizi (2024)[39]	Iran	RCT	E: 27	15–30	E: LLLT	24 h: E: 2.59 ± 2.58/C: 2.04 ± 2.42
				C: 27		C: Ibuprofen	
7	Bernhardt (2001)[40]	America	RCT	E: 13	Average 12	E: Ibuprofen	6 h: E: 2.15 ± 1.12/C: 2.8 ± 1.75
				C: 14		C: Control	24 h: E: 3.05 ± 1.08/C: 3.38 ± 1.35
8	Elshehaby (2023)[41]	Egypt	RCT	E: 25	12–25	E: Ibuprofen	24 h: E: 3.28 ± 1.97/C: 4.24 ± 2
				C: 21		C: Control	
9	Eslamian (2014)[42]	Iran	RCT	E: 74	11–32	E: LLLT	6 h: E: 1.1 ± 1.37/C: 1.41 ± 1.58
				C: 74		C: Control	24 h: E: 1.04 ± 1.72/C: 1.42 ± 2.25
10	Farias (2018)[43]	Brazil	RCT	E: 40	18–40	E: LLLT	24 h: E: 1.99 ± 3.33/C: 4.68 ± 3.63
				C: 40		C: Control	
11	Furquim (2015)[44]	Brazil	RCT	E: 44	13–34	E: LLLT	6 h: E: 0.31 ± 0.64/C: 0.32 ± 0.61
				C: 44		C: Control	24 h: E: 0.73 ± 1.43/C: 0.85 ± 1.71
12	Ganesh (2019)[45]	India	RCT	E: 48	13–20	E: LLLT	24 h: E: 2.89 ± 1.17/C: 7.31 ± 1.76
				C: 48		C: Control	
13	Hosseinzadeh (2016)[46]	Iran	RCT	E: 29	average 15	E: Ibuprofen	6 h: E: 1.05 ± 1.01/C: 2.88 ± 2.08
				C: 28		C: Control	24 h: E: 2.53 ± 1.71/C: 5.83 ± 2.7
14	Kaya (2021)[47]	Turkey	RCT	E: 20	>18	E: LLLT	6 h: E: 2.1 ± 2.02/C: 1.5 ± 1.53
				C: 20		C: Ibuprofen	24 h: E: 2.5 ± 1.52/C: 2.0 ± 1.02
15	Kim (2013)[48]	South Korea	RCT	E: 28	Average 22.7	E: LLLT	6 h: E: 1.96 ± 0.55/C: 4.74 ± 0.5
				C: 30		C: Control	24 h: E: 2.66 ± 0.63/C: 5.55 ± 0.56
16	Kohli (2011)[49]	India	RCT	E: 30	13–20	E: Ibuprofen	6 h: E: 4.69 ± 3.14/C: 5.78 ± 2.96
				C: 30		C: Control	24 h: E: 5.32 ± 1.87/C: 6.66 ± 2.19
17	Marini (2015)[50]	Italy	RCT	E: 30	20–25	E: LLLT	24 h: E: 0.63 ± 1.03/C: 3.11 ± 2.72
				C: 30		C: Control	
18	Minor (2009)[51]	America	RCT	E: 16	13–30	E: Ibuprofen	6 h: E: 4.1 ± 1.3/C: 8.5 ± 1.3
				C: 18		C: Control	
19	Mirhashemi (2021)[52]	Iran	RCT	E: 30	13–37	E: LLLT	6 h: E: 6.81 ± 1.99/C: 7.27 ± 1.99
				C: 30		C: Control	24 h: E: 7.22 ± 1.95/C: 7.92 ± 1.85
20	Nóbrega (2013)[53]	Brazil	RCT	E: 30	>12	E: LLLT	6 h: E: 0.55 ± 1.67/C: 3.11 ± 2.45
				C: 30		C: Control	24 h: E: 1.56 ± 2.32/C: 5.03 ± 3.34
21	Patel (2011)[54]	America	RCT	E: 18	18–30	E: Ibuprofen	6 h: E: 0.78 ± 0.95/C: 1.35 ± 1.28
				C: 18		C: Control	24 h: E: 2.31 ± 1.53/C: 2.94 ± 1.77
22	Rumão (2020)[55]	Brazil	RCT	E: 30	Average 28	E: LLLT	24 h: E: 2.19 ± 2.09/C: 4.25 ± 2.49
				C: 30		C: Control	
23	Sudhakar (2014)[56]	India	RCT	E: 39	14–21	E: Ibuprofen	6 h: E: 1.57 ± 1.21/C: 5.35 ± 4.98
				C: 41		C: Control	24 h: E: 2.11 ± 1.67/C: 4.47 ± 2.74

C = control group, E = experimental group, LLLT = low-level laser therapy, RCT = randomized controlled trial, VAS = visual analog scale.

The NMA structures encompassed 3 direct comparisons: "LLLT vs Control," "Ibuprofen vs Control," and "LLLT vs Ibuprofen." Notably, both network graph of 6 and 24-hour indicated that the comparison between the LLLT group and the control group encompassed the largest number of studies (Fig. 3A and B). More details can be found in contribution plot (Fig. S4A and B, Supplemental Digital Content, https://links.lww.com/MD/P534). Global and local inconsistency tests were used to assess inconsistencies, and both 6-hour and 24-hour indicated no significant inconsistency ($P > .05$) (Table S5, Supplemental Digital Content, https://links.lww.com/MD/P535).

According to the pairwise forest plot, LLLT shows a significant reduction in pain at both 6 and 24-hour after the placement of elastomeric separators when compared to the control group, with MDs and 95% confidence intervals of −1.46 (−2.28 to −0.64) and −1.84 (−2.52 to −1.16), respectively. Similarly, Ibuprofen therapy exhibits a notable efficacy in alleviating pain at 6- and 24-hour post-placement, compared to the control group, with MDs and 95% confidence intervals of −1.93 (−2.93 to −0.93) and −1.63 (−2.58 to −0.68), respectively (Fig. 4A and B).

To rank the viability and reveal the probabilities of effectiveness ranking among the entire treatment procedures, the rankograms and the SUCRA were delivered in NMA structures. In the 6-hour of NMA, the Ibuprofen therapy appeared to be the greatest probability (76.9%) to become the optimal intervention. It also had the highest mean rank (1.2) in the rankograms and the biggest value of SUCRA (88.4). Also, the LLLT therapy (SUCRA: 61.5) works better than control groups, but not as well as Ibuprofen, so it had the greatest probability (23.1%) of being second place in the rank. (Fig. 5A) In the 24-hour of NMA, the LLLT therapy showed the greatest probability (63.6%) to be the best intervention. It also had the biggest value of SUCRA (81.8). Meanwhile, the ibuprofen therapy works better than control groups (SUCRA:68.2), but not as well as LLLT, so it had the greatest probability (36.4%) of being second place in the rank (Fig. 5B).

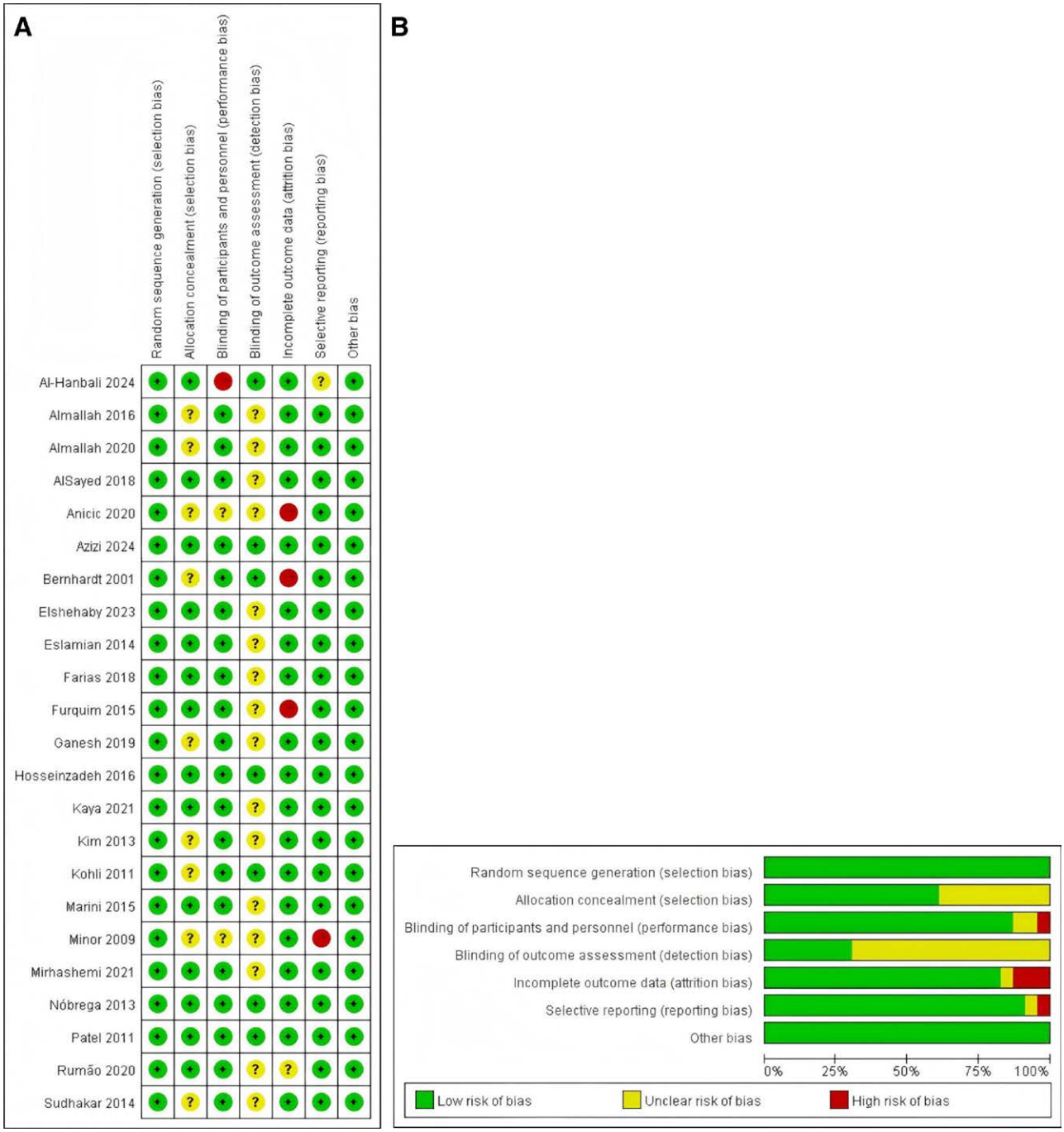


Figure 2. (A) Risk of bias summary. (B) Risk of bias graph.

Table 2

Traditional meta-analysis results of different interventions.

		Heterogeneity			Meta-analysis results		
Time point	**Interventions**	I^2	***P***	**Effect model**	**MD**	**95% CI (%)**	***P***
6-h	LLLT	97%	<.00001	Random-effects model	−1.44	(−2.42, −0.46)	.004
	Ibuprofen	90%	<.00001	Random-effects model	−1.88	(−3.07, −0.70)	.002
24-h	LLLT	93%	<.00001	Random-effects model	−1.89	(−2.71, −0.06)	<.00001
	Ibuprofen	72%	.0009	Random-effects model	−1.43	(−2.17, −0.69)	.0001

CI = confidence interval, I^2 = *I*-squared, LLLT = low-level laser therapy, MD = mean difference, *P* = *P*-value.

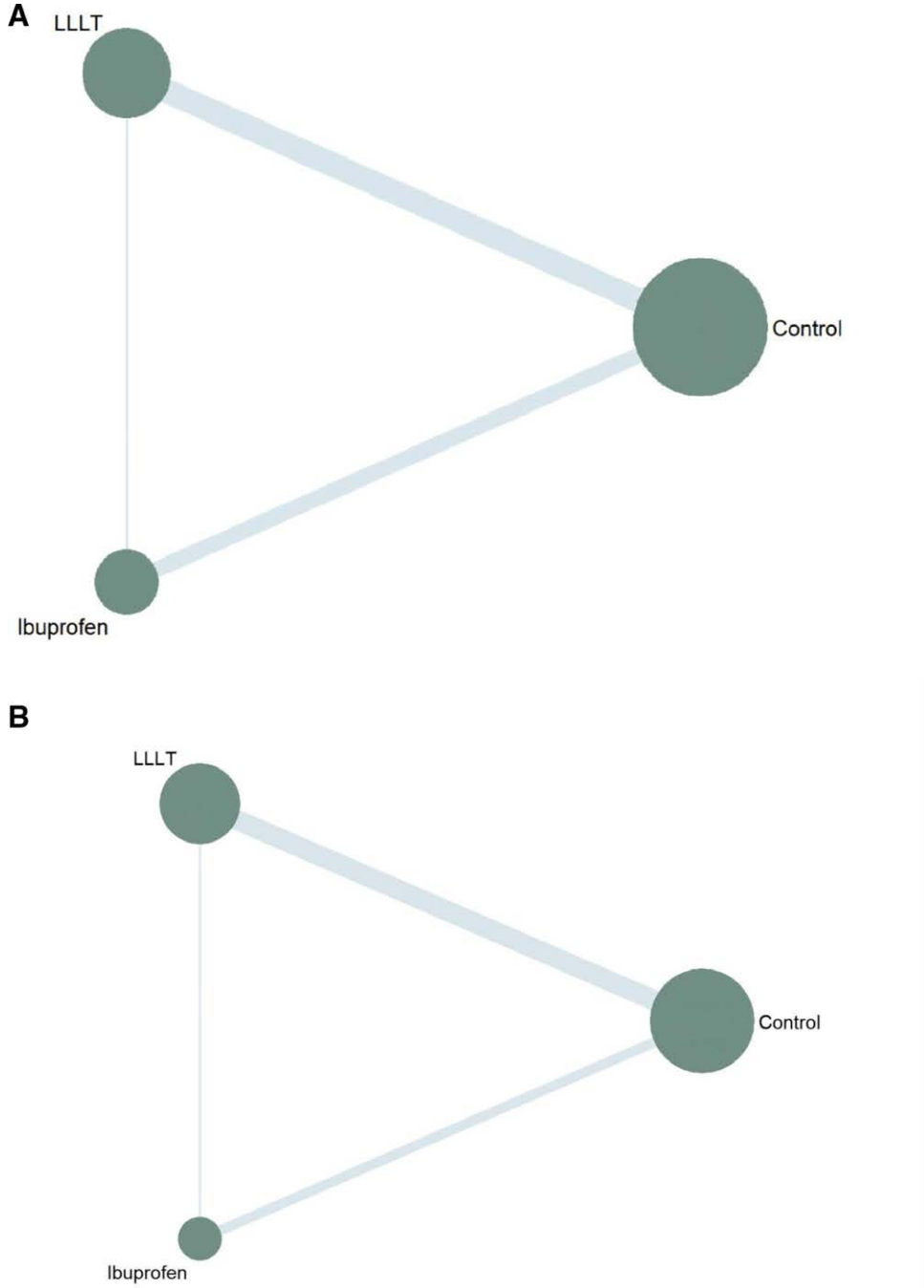


Figure 3. Network graphs for various outcomes. (A) Outcomes at 6-h after orthodontic elastomeric separator placement; (B) outcomes at 24-h after orthodontic elastomeric separator placement.

In sensitivity analyses, the effect value and conclusion were slightly altered after excluding the study with a high risk-of-bias, but the ranking of treatments in this NMA remained unchanged (Fig. S5A and B, Supplemental Digital Content, https://links.lww.com/MD/P534). Furthermore, funnel plots (Fig. S6A and B, Supplemental Digital Content, https://links.lww.com/MD/P534) showed no apparent asymmetry, suggesting a low risk of publication bias.

4. Discussion

The early onset and peak intensity of orthodontic pain present significant challenges in managing orthodontic discomfort, thereby affecting patient compliance and contributing to their psychological distress.[27] This study evaluated the analgesic efficacy of ibuprofen and LLLT at 2 key time points – 6 and 24 hours after separator placement – using VAS scores. Our findings indicate that both interventions significantly reduced pain compared with control, with ibuprofen demonstrating greater pain relief at 6 hours and LLLT showing more sustained effects at 24 hours. These results align with previous systematic reviews and meta-analyses on orthodontic pain management. Prior studies have shown that LLLT effectively reduces separator-induced pain, particularly within the first 24 hours.[57,58] Similarly, Cheng et al found that ibuprofen has a rapid onset of action, with peak

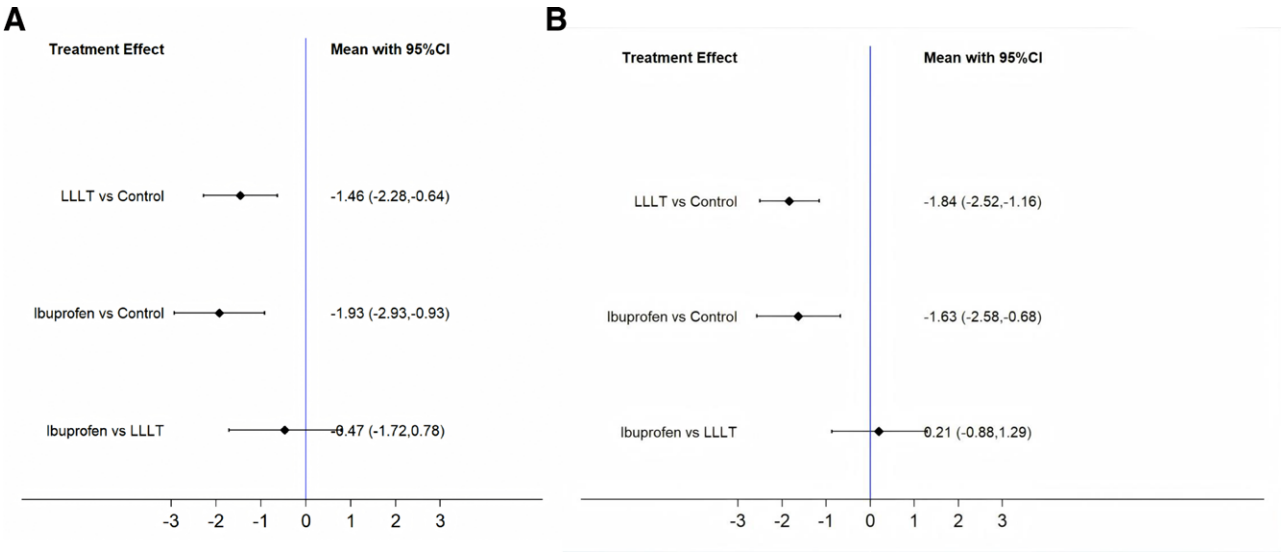


Figure 4. Pairwise forest plot for (A) 6-h and (B) 24-h after orthodontic elastomeric separator placement.

A

Treatment	SUCRA	PrBest	MeanRank
Control	0.0	0.0	3.0
LLLT	61.5	23.1	1.8
Ibuprofen	88.4	76.9	1.2

B

Treatment	SUCRA	PrBest	MeanRank
Control	0.0	0.0	3.0
LLLT	81.8	63.6	1.4
Ibuprofen	68.2	36.4	1.6

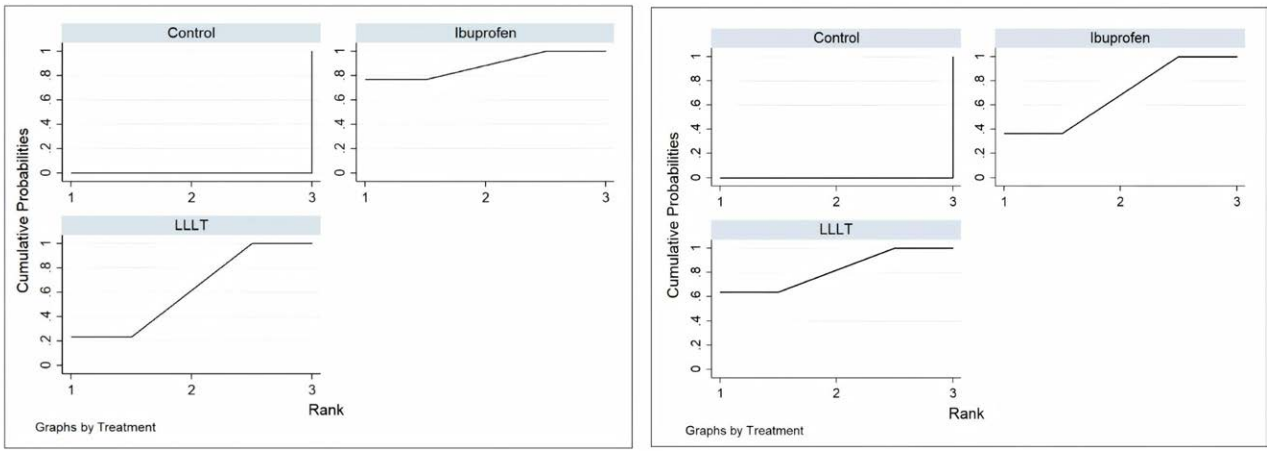


Figure 5. The rankograms and SUCRA in (A) 6-h and (B) 24-h after orthodontic elastomeric separator placement. SUCRA = surface under the cumulative ranking.

analgesic effects observed at 6 hours and a gradual decline by 24 hours.[59] These patterns underscore the distinct time-dependent analgesic profiles of the 2 interventions. Moreover, the NMA conducted by Al-Hanbali et al (2024) further supports the complementary nature of ibuprofen and LLLT, showing that ibuprofen offers immediate pain control while LLLT offers increasing pain reduction over time. Collectively, these findings support the potential of a time-adjusted, multimodal pain management strategy – administering preemptive ibuprofen for early relief and LLLT for sustained control – may be an effective approach to enhance patient comfort, adherence, and overall treatment experience in orthodontics.

The observed differences in analgesic timing likely reflect the distinct mechanisms of action of the 2 intervention. Orthodontic force application triggers acute inflammation in the periodontal ligament, involving vascular dilation, prostaglandin release, and nociceptor activation.[60] NSAIDs like ibuprofen provide early-phase relief by inhibiting cyclooxygenase enzymes, thereby reducing prostaglandin synthesis and associated inflammation.[61] However, concerns regarding their impact on osteoclast activity and tooth movement, as well as potential gastrointestinal and renal or cardiovascular side effects, underscore the need for short-term, targeted use.[62–65] In contrast, LLLT achieves its effects via photobiomodulation, enhancing ATP production, tissue repair, and microcirculation.[66] It modulates inflammatory markers and inhibits pain transmission pathways by downregulating TRPV1 and CGRP expression.[67,68] In orthodontics, LLLT modulates osteoblast and osteoclast activity to accelerate tooth movement and bone remodeling.[69] These downstream effects contribute to its more gradual but sustained analgesic benefits, aligning with pain progression in later phases.[70]

Despite protocol variability across studies, some practical patterns emerged. Most ibuprofen studies employed a single 400 mg dose administered approximately 1 hour prior to separator

placement, supporting the effectiveness of preemptive pharmacological analgesia. For LLLT, although parameters varied, most protocols applied a single session immediately post-separation, with wavelengths between 808 and 830 nm and energy densities ranging from 1 to 4 J/cm^2. These trends suggest that both pharmacological and non-pharmacological interventions, when applied in the early-phase of pain onset, can be clinically useful in attenuating peak orthodontic discomfort. However, further high-quality randomized clinical trials are needed to explore the synergistic potential of combined or time-staggered interventions and to identify optimal dosing regimens for each modality.

When interpreting the results of this study, it is essential to consider several limitations. First, Only English-language studies were included in this review, which may introduce a degree of language bias. Second, gray literature such as conference abstracts and unpublished trials was not included, potentially contributing to publication bias. Third, to further explore potential sources of heterogeneity, we attempted subgroup analyses based on participant characteristics such as age and sex. However, these analyses were limited by the small number of eligible studies. As a result, more in-depth subgroup analyses could not be performed. Additionally, due to the lack of access to individual-level data, it was not possible to retrieve the original distributional characteristics of pain scores. This may have introduced a degree of imprecision in the pooled effect estimates. Finally, heterogeneity in intervention protocols – particularly regarding the timing, dosage, and frequency of ibuprofen and LLLT – may compromise the reliability of treatment rankings derived from SUCRA and PrBest values. These rankings should be interpreted cautiously and contextualized alongside MDs and confidence intervals. Moreover, the robustness of ranking results may be influenced by several factors, including small sample sizes in some treatment arms, risk-of-bias in certain included studies, and sparsity in parts of the treatment network.[71] Future research should focus on large-scale, well-controlled RCTs with standardized intervention parameters and comprehensive patient profiling. Moreover, direct comparisons of sequential or combined ibuprofen and LLLT protocols may yield further insights into the efficacy of personalized, time-sensitive pain management strategies in orthodontic care.

5. Conclusion

Both ibuprofen and LLLT were found to be effective in reducing pain following the placement of orthodontic elastomeric separators. Ibuprofen provides more immediate analgesic effects, while LLLT offers more sustained pain relief over a 24-hour period. These findings highlight the distinct time-dependent mechanisms of the 2 interventions and support a flexible, evidence-based approach to orthodontic pain management. Clinicians may consider tailoring treatment strategies according to the timing and severity of discomfort, as well as patient-specific factors such as medication tolerance and clinical goals. To further guide clinical decision-making – particularly for specific patient subgroups with limited evidence – future research should prioritize high-quality, head-to-head RCTs comparing standardized protocols, as well as studies evaluating the combined or sequential use of ibuprofen and LLLT as a multimodal, time-adjusted strategy.

Author contributions

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Supervision: Shasha Gao, Yang Cheng.
Validation: Shasha Gao, Peipei Wu.
Visualization: Shasha Gao, Peipei Wu.
Writing – original draft: Shasha Gao, Peipei Wu.
Writing – review & editing: Yang Cheng.

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